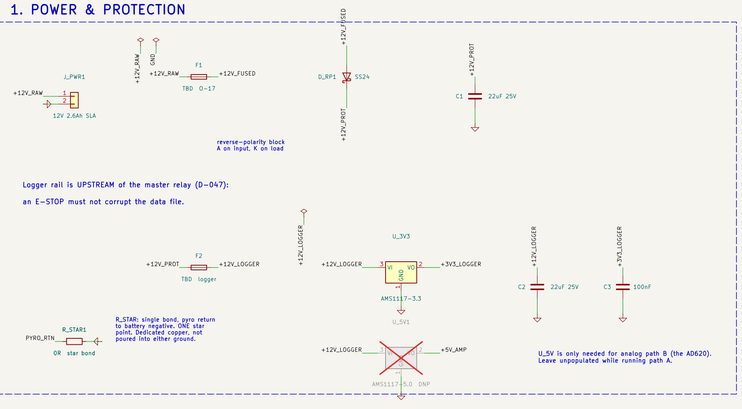
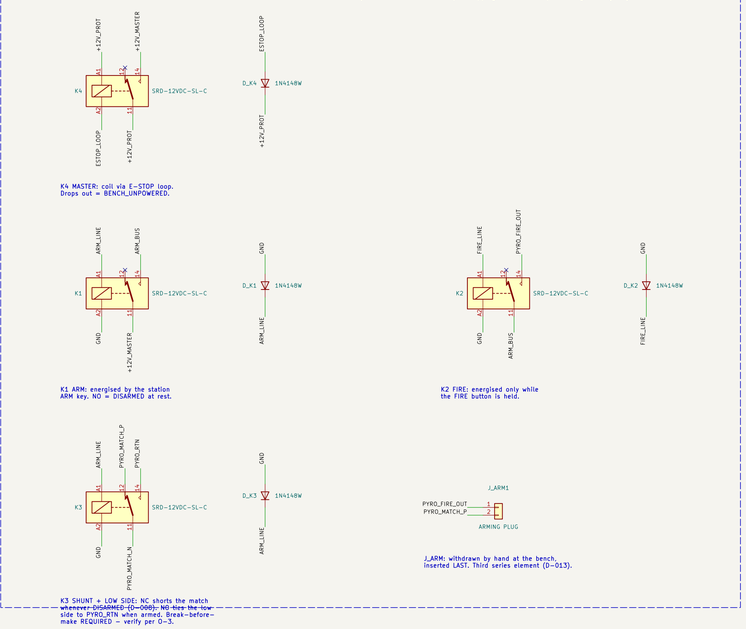
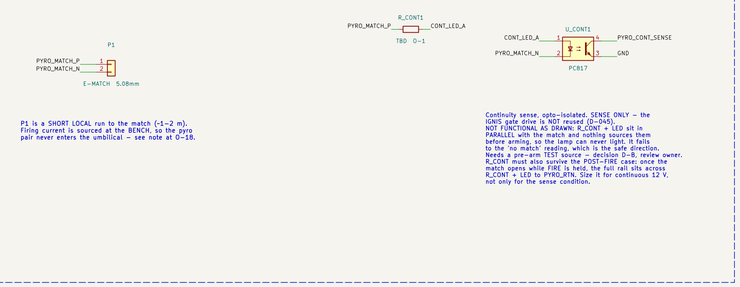
1. POWER & PROTECTION



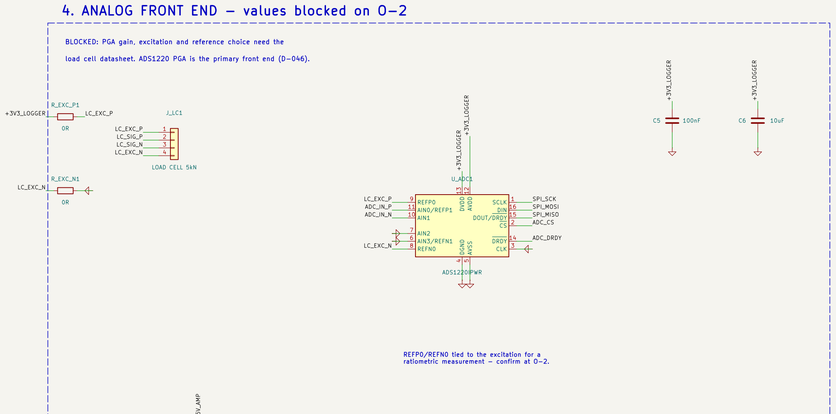
2. SAFETY CHAIN — hardware only, no MCU



3. PYRO INTERFACE — values blocked on O-1

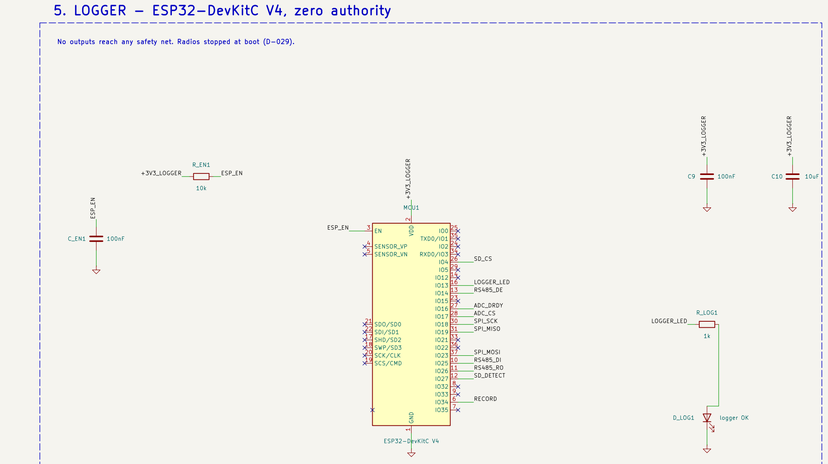


4. ANALOG FRONT END — values blocked on O-2



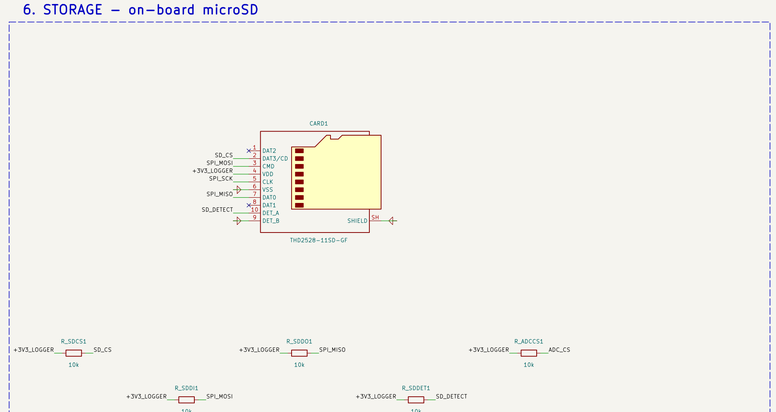
AD620 minimum total supply is 4.6 V, so it CANNOT run on 3.3 V.
Path B therefore needs the +5V_AMP rail, and its output must be kept
inside the ADS1220 input range (D-7). Single-supply use also needs
AMP_REF biased above ground — value TBD with D-2.

5. LOGGER — ESP32-DevKitC V4, zero authority

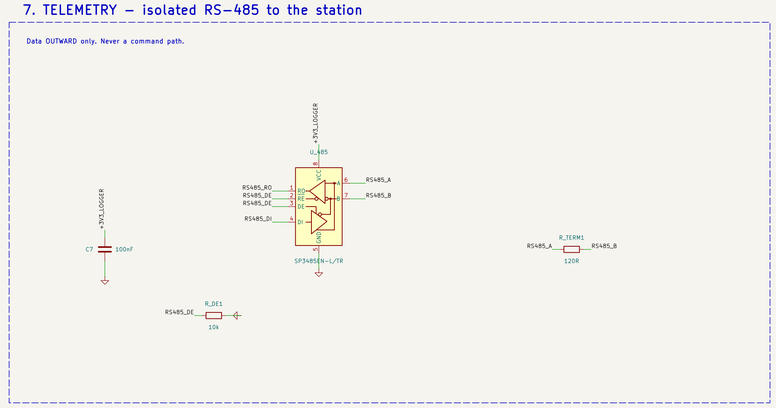


MCU1 is drawn as the ESP32-WROOM-32D module so the GPIO names are readable.
The PHYSICAL part is a socketed ESP32-DevKitC V4 on 2 x PPTC191LFBN-RC (D-042),
so the symbol-to-socket pin mapping is set when the DevKitC footprint is made (D-16).
Pins avoid GPIO6–11 (SPI flash) and all strapping pins 0, 2, 5, 12, 15 (D-031).
RECORD sits on IO34 because 34–39 are INPUT-ONLY and cannot become outputs.

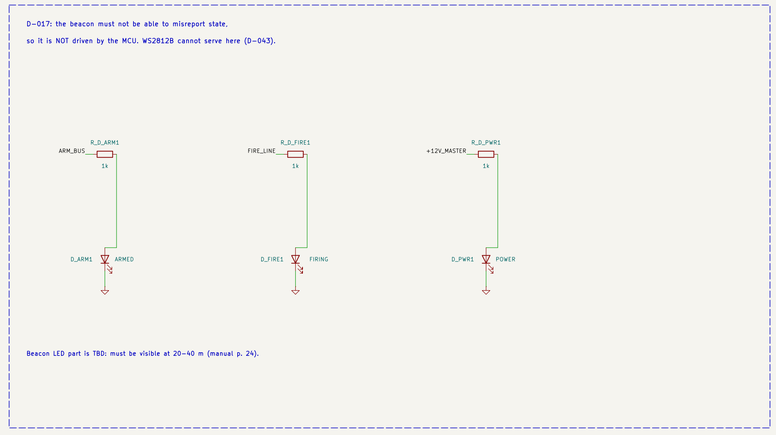
6. STORAGE — on-board microSD



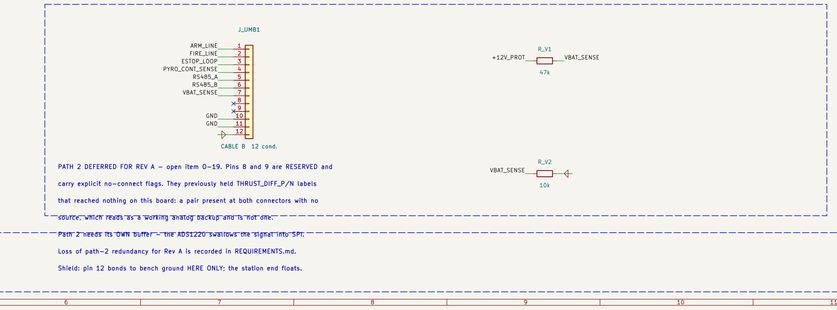
7. TELEMETRY — isolated RS-485 to the station



8. INDICATORS — beacon driven from the REAL arm/fire lines



9. UMBILICAL — cable B, control + analog



TBD values await O-1 (e-match no-fire current) and O-2 (load cell datasheet).

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_bench.py
Igniter safe state is SHORTED via K3 normally-closed. Values marked TBD await O-1 / O-2.
PRELIMINARY — NOT APPROVED. NO MCU IN THE ARM, FIRE OR E-STOP PATH.

Ignitia Rocket Lab	
Sheet: /	
File: talavera-bench.kicad_sch	
Title: TALAVERA v1 — Bench Board	
Size: A1	Date: 2026-11-06
KiCad 10.0.0	Rev: A
	Id: 1/1